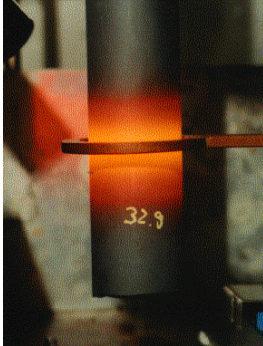




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Technology of Metallic Materials A.Y 2025-26

Mechanical Engineering Course



Thermal and Chemical Surface
Treatment of Steels – Welding
Phenomena

G. Maizza
DISAT

Lecture 10

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Hardening by Carburizing

- Purpose: high hard surface and tough core
- Exploited factors:
 - martensite increases its hardness with increasing %C
 - martensite forms from austenite on quenching
 - C is highly soluble in austenite
 - absorption/ diffusion rate increases < with increasing T
 - Process conditions: **T_c=870-950 °C** **time=10-20 h**
 - Applicable to carburizing steels (plain carbon steel <0.2% and low carbon low alloy steels)
- Applications: gears, shaft, bearings. piston rod, etc.
- Process outcome
 - Carbon concentration profile along the thickness (max HV on surface)
 - Transformation of austenite into martensite by oil quenching

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Hardening by Nitriding

- Purpose: extremely hard surface and tough core
- Exploited factors:
 - N with Fe form very hard compound (nitrides)
 - Solubility of N in ferrite is very low
 - At the stability temperature of ferrite, diffusion is slow
 - the formation of nitrides inhibits further diffusion of N
 - Process conditions: $T_N = 500-600\text{ }^{\circ}\text{C}$ time=70 h
 - Applicable to nitriding steels (plain medium carbon steel <0.4-0.6% and medium carbon low alloy steels in tempered/semi-finished conditions)
- Applications: gears, valves, injection pump parts, high wear parts, high localized loaded parts
- Process outcome:
 - Formation of very high hardness superficial nitrides ~ 1100 HV
 - 0.1-0.6 mm penetration depth;

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Hardening by Nitriding

0.2-0.45% 1%

Steels 0.2-0.5%C + Al, Cr, V, Mo

↓

500°C, 40-100 h

↓

$2\text{NH}_3 \rightleftharpoons 2\text{N} + 3\text{H}_2$

↓

Atomic N dissolves into the steel and

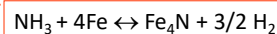
↓

strato esterno: $(\text{Fe,Al,Cr})_2\text{N}$
strato più interno: $(\text{Fe,Al,Cr})_4\text{N}$

γ' Fe_4N



Thin nitrided layer (0.2-0.3 mm)



γ' very hard interstitial compound up to 6% N in a restricted compositional range

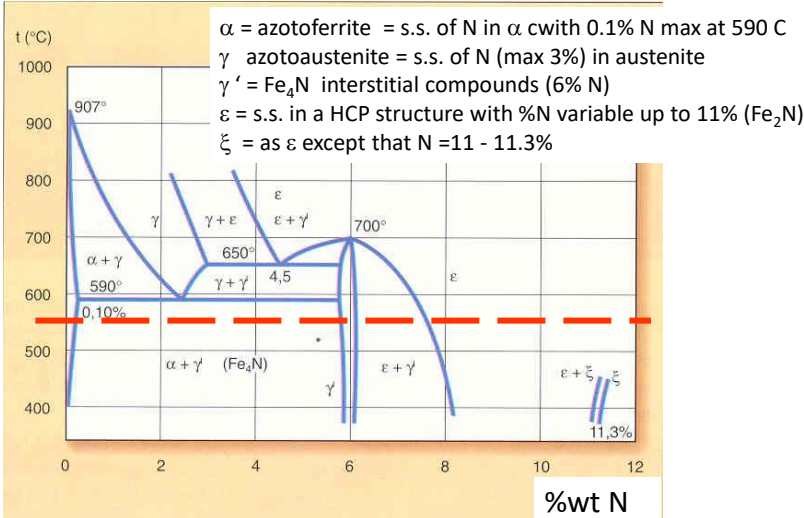
The gas is composed by partially dissociated NH_3 ammonia to prevent the formation of inert N_2 and excessive surface enrichment ($\rightarrow \epsilon, \zeta$); N can diffuse in Fe, at atmospheric pressure
At $T = 400 - 700\text{ }^{\circ}\text{C}$ NH_3 is metastable in presence of N and H

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Fe-N METASTABLE PHASE DIAGRAM

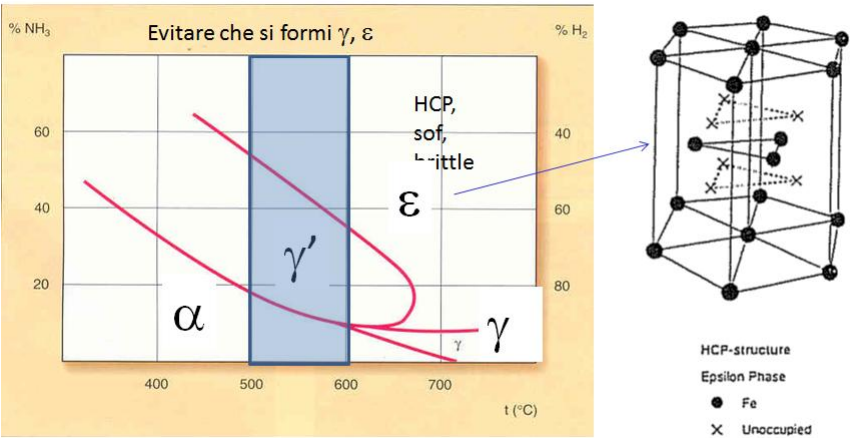


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Fe-N METASTABLE PHASE DIAGRAM



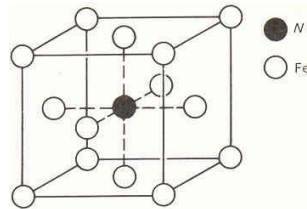
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NITRIDING STEELS

Sigla UNI	30 Cr Mo 10	42 Cr Al Mo 7	38 Cr Al Mo 7
% C	0,30	0,42	0,38
% Si	0,30	0,35	0,35
% Mn	0,60	0,55	0,60
% Cr	2,50	1,70	1,70
% Mo	0,40	0,25	0,35
% Al	—	0,40	1,0
Hardness after nitriding	650 HV	900 HV	1050 HV

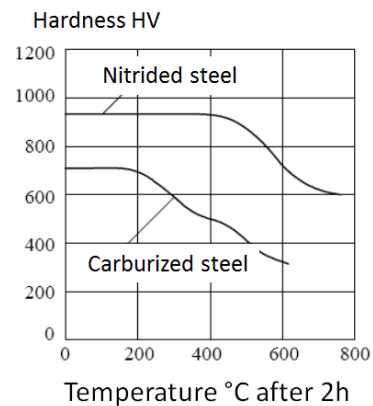
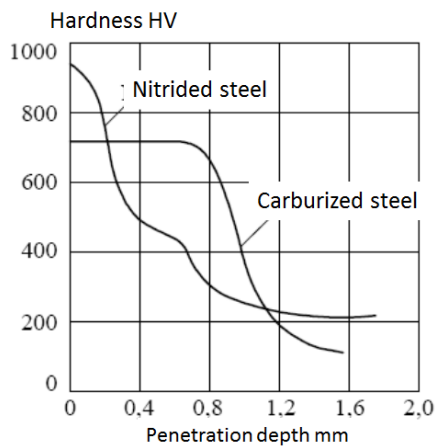


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Carburizing vs Nitriding



Harris' formula

$$\text{Penetration depth (mm)} = 660 \cdot \sqrt{t} \cdot \exp\left(-\frac{8287}{T}\right)$$

T [K]
 t [h]

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Heat Treatments for post-carburized and post-nitrided steel parts

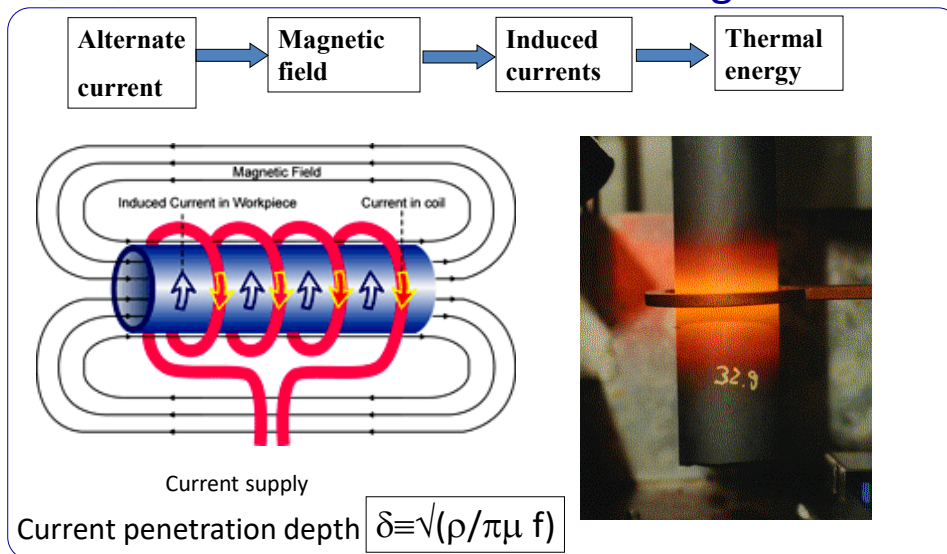
Carburized	Nitrided
Normalizing +	Normalizing +
work hardening annealing	Work hardening annealing
Machining roughing (medium thickness)	Machining roughing (medium thickness)
Carburizing	Tempering
Quenching + tempering at 150 °C	Finish Nitriding
Finish by grinding	Finish by grinding

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Basic principle of Induction hardening

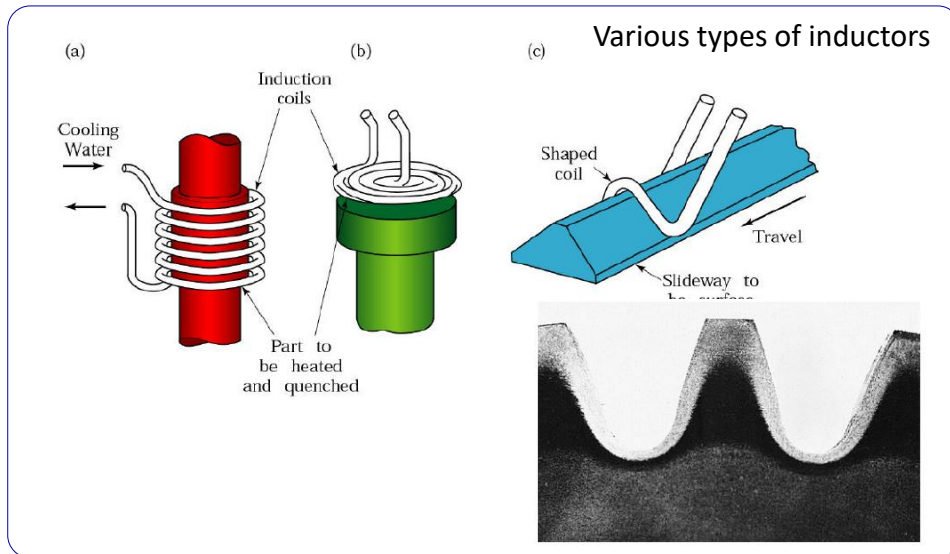


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Induction Thermal Hardening

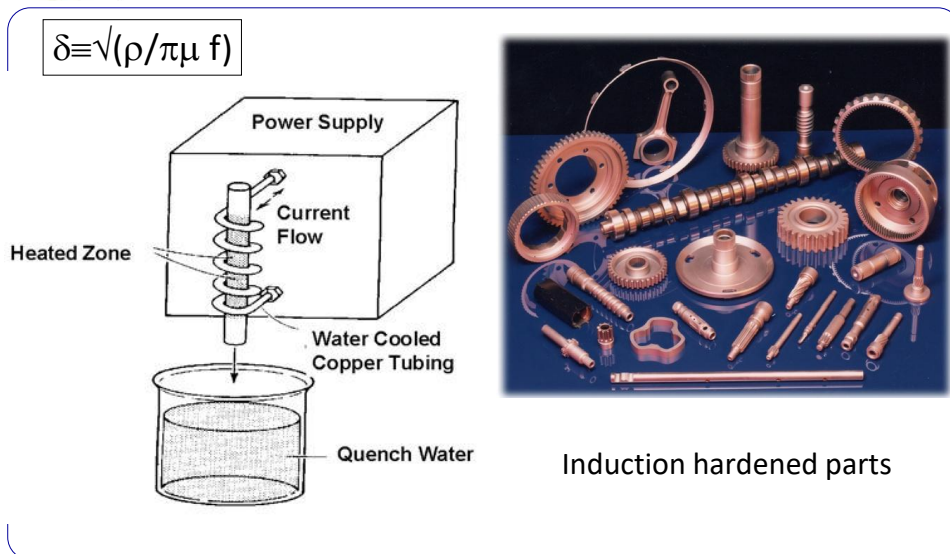


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Induction hardening (single-shot)

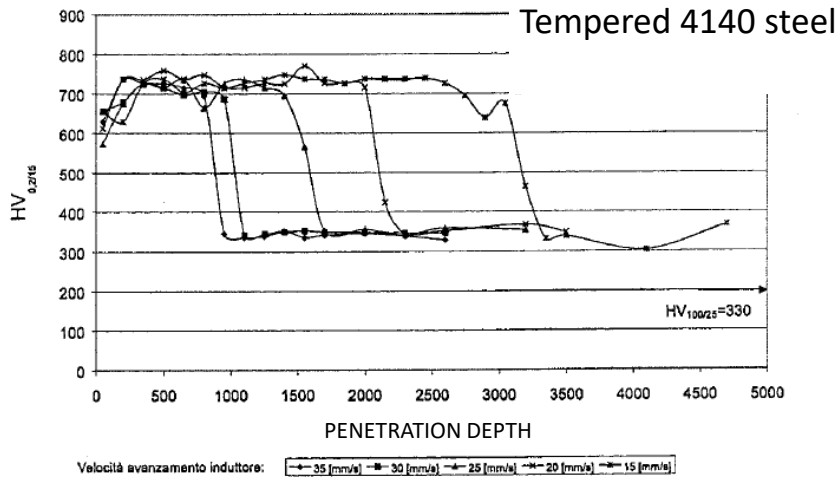


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Induction heat treatment: hardness profiles along the thickness



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Heat vs. Chemical Surface Treatments

PROCESS	STEELS	COMPONENTS	CONDITIONS
INDUCTION 3 kHz	0.38wt. % C	Axle	Water Quenched. Tempered at 210°C
FLAME HARDENING	0.67wt. % C	Wheel	Water Quenched. Tempered at 490°C
CARBURINZING	0.2wt. % C-1.8wt. % Ni -0.5wt. % Cr-0.2wt. % Mo	Gear	Oil Quenched. Tempered at 200°C
Laser 15 kW	0.43wt. % C-1.5wt. Mn	Gear	Self Quenched
NITRIDING	0.2wt. % C-1wt. % Cr -0.1wt. % V-0.08wt. % Al	Gear	Process Carried out at 570°C

(c)

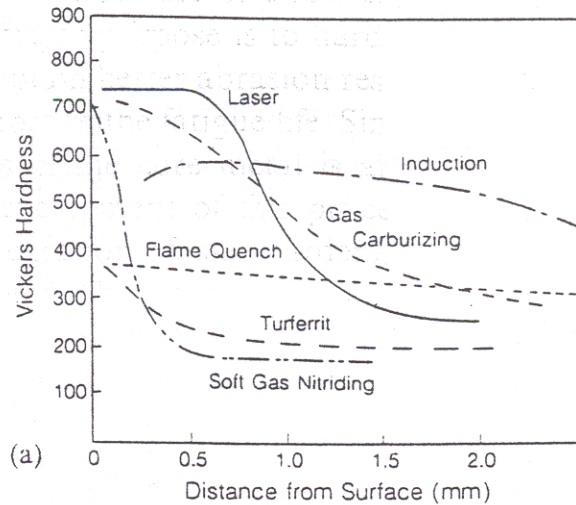
These surface heat treatments, beside originating a high surface hardness, will determine an increase in fatigue resistance as all surface heat treatments will confer a compressive stress state on the surface.

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Heat vs. Chemical Surface Treatments

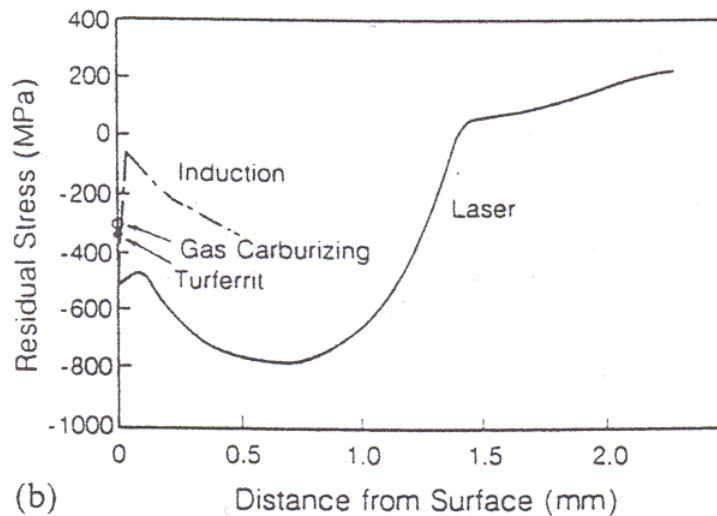


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Surface Heat vs. Chemical Surface Treatments

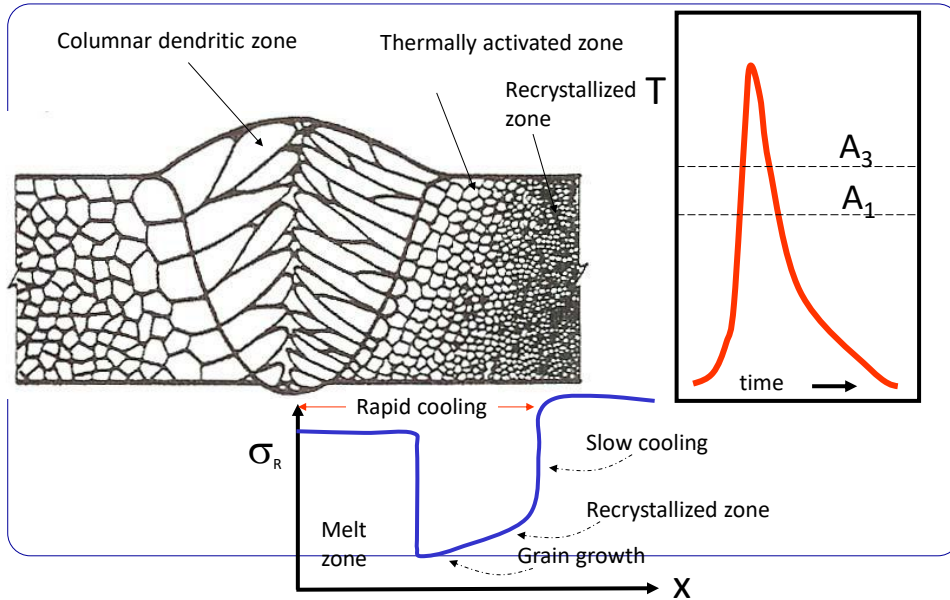


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Weldability of Steels



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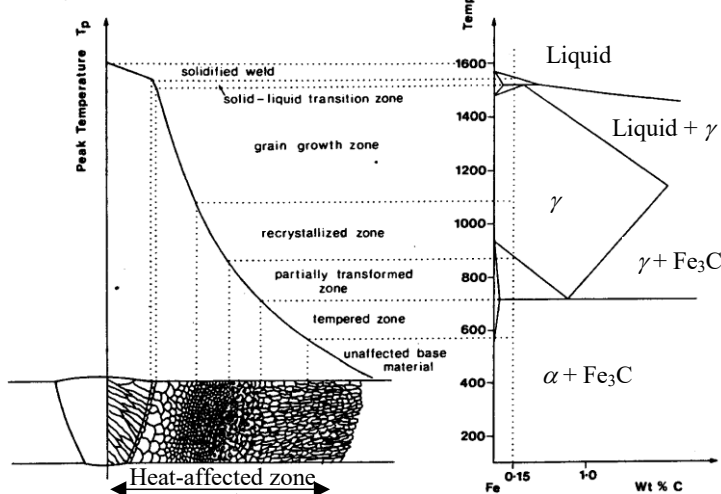


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Weldability of steels: HAZ

Welding of hypoeutectoid steel (0.15 wt% C) vs Fe-Fe₃C diagram

(From K. Easterling, *Introduction to the Physical Metallurgy of Welding*, Butterworths, (1983))

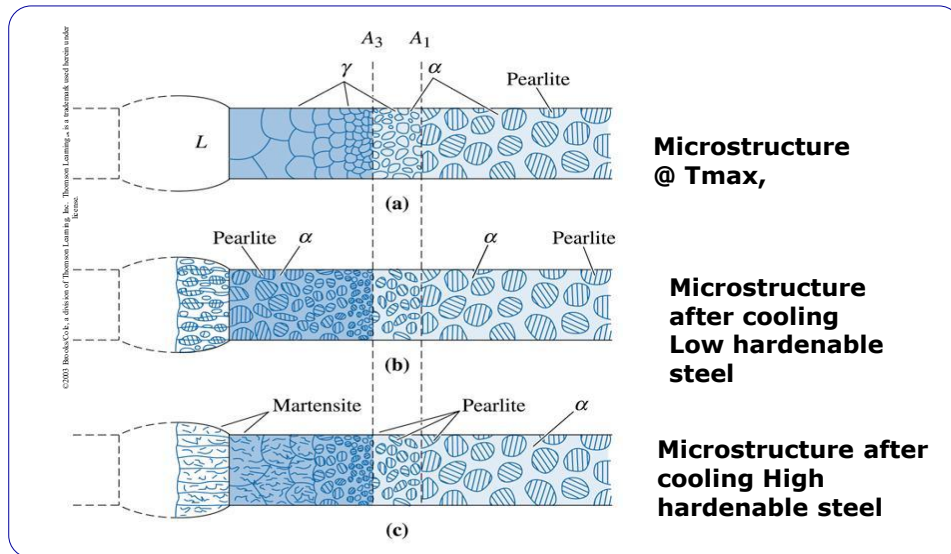


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Formation of HAZ

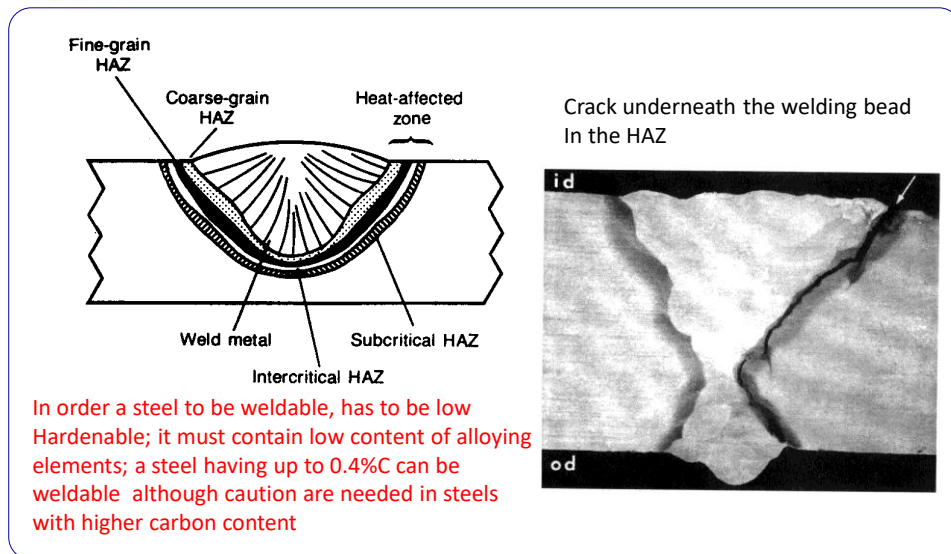


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Microstructure in the welding zone: low carbon steels



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Chemical Surface Treatment: Summary

Process	Nitriding	Nitro-carb.	Boriding	Carburizing	Process QUAD
Superficial compound	Fe-N	Fe-(C,N)	Fe-B	Fe-C	CrCN, VCN, TiCN
T sup. °C	500-550	550-650	600-1000	700-1050	400-700
T core °C	500-550	550-650	600-1000	700-1050	400-700
Proc.time(h)	1-70	1-6	1-4	1-46	1-8
Thickness µm	50-100	50-200	50-500	100-7000	5-20
hardnessHV	700-1000	700-1000	1000-1800	750-950	1200-2500
Distorsion	min	min	high	high	min
Initial H.T.	yes	yes	after	after	yes
Post-trattam.	seldom	seldom	seldom	needed	seldom
Unif. Spess.	good	good	good	good	good

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WEAR RESISTANT CERAMIC COATINGS

Process	Cr hard	Thermal CVD	Plasma CVD	PVD	Flame coating	Stellite
Surface layer	Cr	TiC, TiCN, TiN	TiC, TiN	TiN, CrN	Ni, Cr, B, Si	Ni, Cr, B, Si
Method	electrolysis	gas	Arc in vacuum	N in vacuum	Powder melting in plasma	melting (arc)
Surface T°C	50-80	800-1100	300-600	300-600	1000-1100	≈ T _{fus}
Core	50-80	800-1100	300-600	300-600	500-800	300-900
Time (h)	1-6	4-8	1-3	2-4	dep. n size	dep. on size
thickness	20-50	3-15	1-6	1-6	500-2000	2000-5000
HV hardness	700-800	1500-2000	1500-2000	2000-4000	600-800	300-900
Distorsion	low	high	low	low	high	high
Substrate	before	after	before	prima	before or after	Before or after
Post-treatment.	seldom	seldom	seldom	seldom	needed	needed
Uniformity	low	good	poor	poor	Very poor	Very poor
Substrate.	Met./non	Metals	Metals	Metals	Metal/nonMet	Metals

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